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Integrating perceptual cues with mixture-of-experts for low-light image restoration

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ABSTRACT

Visual perception of nighttime images is often impaired by degradations: low-light, noise, motion blur, and low-resolution. While recent methods have made progress in jointly solving these degradations, the diversity of patterns and intensities in degradation has not been properly considered, leading to inconsistent illumination and unintended artifacts. In response, we propose to integrate perceptual cues with mixture-of-experts (IPCMoE) to achieve flexible processing for low-light low-quality images. By exploiting the perceptual cues, we strategically combine dedicated experts with the selective collaboration approach for feature enlightening and texture restoration. To this end, we develop perceptual-integrated MoEs by designing customized routers and task-dependent experts. Specifically, the texture memorial MoE is developed to preserve valuable features to restore high-fidelity details, and the enhancement MoE that adaptively integrates enlightening cues and texture cues is designed to formulate the relationship between feature enlightening and texture restoration, thereby achieving dynamic image processing. Compared to state-of-the-art models, our IPCMoE achieves superior performance on various benchmarks for handling complex low-light scenes.

1. Introduction

Increasing the visibility of nighttime scenes gains considerable interest in the image processing community (He et al., 2025a; Ren et al., 2025). As the real-world image degradations are usually much more complex. In addition to the low-light degradation, noise (He et al., 2025a), motion blur (Yuan et al., 2007), and low-resolution are also common problems in nighttime photography, leading to texture smear, edge distortion, and insuffcient details of the captured images. Therefore, simultaneously handling visual degradations is significant for improving the perception of nighttime images.

Unfortunately, despite numerous low-light enhancement (LLE) methods and image restoration methods (i.e., model for deblurring or super-resolution) proposed, combining them is not a practical attempt to recover the co-exist degradations: On the one hand, sequentially adopting the existing techniques may increase the likelihood of error accumulation (Zhou et al., 2022), thus introducing unintended artifacts; On the other hand, the joint retraining approach suffers from the increased computational costs and the inherent gap between models (Li et al., 2024c).

In light of the existing methods for couple-degraded low-light image restoration, the following pioneering attempts have been made. For joint low-light enhancement and deblurring: Zhou et al. (2022) propose the LEDNet with an encoder for enlightening and a decoder for deblurring, respectively. Zou et al. (2024) introduce the vector quantization codebook to match the illumination priors with degraded features for structure and detail restoration. Li et al. (2024c) solve the joint problem with a parallel multi-task learning architecture by designing two perceptual auxiliary tasks. For joint low-light enhancement and super-resolution: Rasheed and Shi (2022) first enhance dark features in low-resolution spaces and then perform upsampling. Xu et al. (2023) propose adjusting the relative underexposure levels to a uniform level to avoid artifacts, and introduce the a strategy for sampling multi-scale patches. Ye et al. (2024a) develop a semantic prior guided interactive network to produce normal-light high-resolution images. In summary, the key ingredient of the existing approaches is to develop specialized strategies for each type of degradation within an end-to-end framework.

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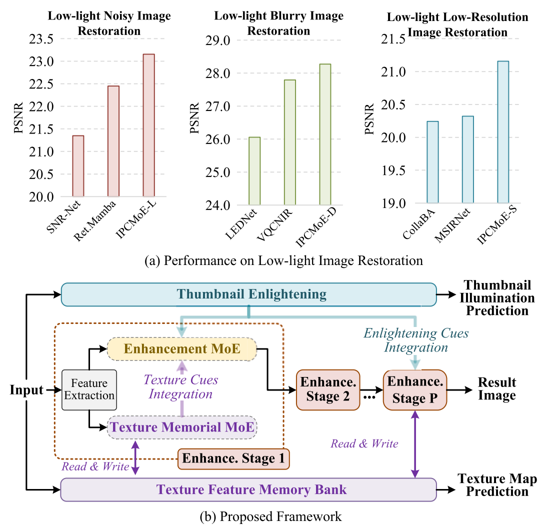


Fig. 1. (a) Comparison on multiple benchmarks shows that the proposed method achieves superior results under complex low-light degradations. (b) Overview of the proposed framework, highlighting the main pipeline and key modules.

Based on this, our work goes further by focusing on the flexibility and adaptability of dedicated strategies in complex scenarios.

We consider that the following difficulties still challenge the existing methods for low-light image restoration: ***Firstly***, degradations of couple-degraded images show intricate patterns and variable intensities. Concretely, the global brightness intensity typically varies from one image to another, and such variation demands adaptive processing for enlightening. Similarly, blurry at different magnitudes should be properly handled for texture restoration. ***Secondly***, the primary factor that impacts image quality may change during the image enhancement process. Specifically, blur degradation is imperceptible in poor illumination conditions, and low-light degradation is currently the main factor affecting image quality. Conversely, when brightness is boosted to an acceptable level, noticeable edge blurry or insuffcient details can greatly affect the visual perception of image textures. Based on these observations, we consider that the relationship between feature enlightening and texture restoration should be formulated to generate visually appealing results.

While seeking solutions to the aforementioned challenges, we realized that the capability of dynamically and adaptively processing the image is an indispensable element. As a result, we naturally turn to the prevalent Mixture-of-Experts (MoE) (Chen et al., 2022b; Riquelme et al., 2021; Ye & Xu, 2023), which provides efficient computation by distributing the model capacity across a set of dedicated experts, while assuming that only a subset of experts is used for each input (Chen et al., 2022b; Riquelme et al., 2021; Ye & Xu, 2023). Our motivation arises from the observation that image degradations often appear in diverse patterns and intensities. However, attempting to capture and restore all degradations with a single, monolithic model can be challenging. In light of the benefits of the MoE, selective expert collaboration enables the network to specialize in different patterns and properties of the image, which facilitates modeling the dependencies between low-light, low-quality images and normal-light, high-quality images (Fig. 1 (a)).

In this paper, we propose to **i**ntegrate **p**erceptual **c**ues with **m**ixture-**o**f-**e**xperts (IPCMoE) for low-light image restoration. As depicted in Fig. 1 (b), we handle low-quality images by designing a progressive architecture that consists of a series of enhancement stages. In each stage, the input is parallelly processed by both the enhancement MoE (E-MoE) and the texture memorial MoE (T-MoE) after feature extraction. Since co-existing image degradations should be eliminated simultaneously, we design auxiliary tasks inspired by perceptual decoupling mechanisms (Li et al., 2024c) to ensure the model can focus on impaired perceptual information during optimization. To this end, the texture map prediction task is developed to assist the texture memorial MoE in extracting texture cues, and the thumbnail enlightening task is designed to extract enlightening cues. Finally, under the guidance of enlightening cues and the integration of texture cues, our enhancement MoE enables adaptive feature processing to produce visually appealing results.

Specifically, in contrast with using standard MoEs directly, we develop perceptual-integrated MoEs by designing dedicated routers and task-depended experts. We devise the “texture feature memory bank” to preserve the encoded texture features at different enhancement stages, and accordingly propose a T-MoE for producing texture cues. In addition, we propose the E-MoE to provide adaptive processing with integrated perceptual cues. To this end, a feature guided router is proposed to integrate the enlightening cues to guide the expert selection process. Meanwhile, our E-MoE explicitly formulates the relationship between feature enlightening and texture restoration, where the texture cues are adaptively merged with an additional expert. In comparison to our previous work (Li et al., 2025), we have conducted a thorough investigation into low-light image restoration by creating a series of models, making experimental improvements, providing additional methodological details, conducting a deeper analysis and carrying out broader evaluations.

The main contributions are summarized as follows:

- We propose to integrate perceptual cues with mixture-of-experts for low-light image restoration. Our method provides a dynamic and adaptive approach for handling couple-degraded images.
- We design dedicated routers and experts to build perceptual-integrated MoEs: a texture memorial MoE is proposed to preserve valuable details for texture restoration, and an enhancement MoE is designed for adaptive feature processing with integrated perceptual cues.
- Experimental results demonstrate the superior performance of our IPCMoE across diverse benchmarks for low-light enhancement (LLE) task, joint low-light enhancement and deblurring (LLE-Deblur) task, and the joint low-light enhancement and super-resolution (LLE-SR) task.

2. Related works

2.1. Image restoration

Low-light image enhancement methods (Cui et al., 2022; Li et al., 2024; Wu et al., 2023; Xu et al., 2020, 2024) aim to improve the visual perception of the low-light scenes. For traditional methods, the histogram equalization and Retinex theory (Hao et al., 2020; Xu et al., 2020) are popular techniques for enhancing brightness and color. For learning-based methods, Wu et al. (2023) proposed to integrate the priors knowledge of semantic segmentation into the low-light enhancement model to preserve color consistency and visual details. Wang et al. (2023b) developed a transformer-based method by introducing axis-based multi-head self-attention and cross-layer attention fusion mechanisms. Xu et al. (2024) propose a bilateral interaction strategy for both local and global perception. Hu et al. (2024b) constructed a zero-element mask set from real-world night-time traffic monitoring images. They also proposed a powerful model by designing a joint-attention transformer and a map-wise cross affine coupling flow (Hu et al., 2024a). Xu et al. (2025) proposed an unbalanced point map prior based on the differences in the proportion of RGB values for low-light image enhancement.

Image deblurring methods (Cho et al., 2021; Guo et al., 2025; Kong et al., 2023; Yan et al., 2023) are developed to recover sharp images from motion blur degradation. Traditional methods restore the blurry image by assuming priors on structural information or physical constraints. In recent years, deep learning-based methods have benefited from powerful networks that are driven by data. Tu et al. (2022) proposed the multi-axis gated MLP block and cross-gating MLP block for image restoration. Mao et al. (2023) propose a hybrid frequency-spatial mechanism by performing selective FFT operations in the residual network block. Yan et al. (2023) designed the Transformer block to preserve local information, and global features were then integrated.

Image super-resolution methods aims to improve image clarity and detail by increasing the number of pixels in an image (He et al., 2025b; Zhang et al., 2025). Traditional methods (such as image interpolation, sparse regression) are prone to introducing noise and artefacts, limiting their application scenarios. Recently, Mei et al. (2021) proposed Non-Local Sparse Attention (NLSA), which globally captures texture information by introducing non-local blocks and sparse attention mechanisms. Chen et al. (2023d) proposed a recursive generalised Transformer model, effectively extracting global information through recursive-generalization self-attention and hybrid adaptive integration. Chen et al. (2023b) proposed HAT, combining channel attention and window-based self-attention mechanisms, while introducing overlapping cross-attention modules to enhance interactions between adjacent window features. Cai et al. (2023) proposed HIPA, using hierarchical block partitioning to gradually restore high-resolution images, and introducing new attention mechanisms and multi-receptor field attention modules.

Despite these advancements, the coupled degradation of night-time images (e.g., motion blur and low resolution), which frequently occurs in low-light conditions, remains a significant challenge for current image restoration models.

2.2. Mixture-of-experts (MoE)

Mixture-of-experts model has achieved remarkable success in increasing model capacity with fewer resources. As a pioneered work, Jacobs et al. (1991) propose the MoE as a technique that combines a series of sub-models with conditional computation. Shazeer et al. (2017) designed a sparse MoE layer by selecting a specific number of experts for each example, which enabled the training of thousands of sparse experts in a model. Puigcerver et al. (2023) designed a fully-differentiable sparse Transformer by performing a soft assignment of all input tokens to each expert.

By enabling automatic conditional computation for each sample, MoE techniques (Jacobs et al., 1991; Ruiz et al., 2021; Shazeer et al., 2017) have shown promising performance in natural language processing, vision-language learning task (Bao et al., 2022), multi-task learning (Chen et al., 2023c; Ye & Xu, 2023), and high-level vision tasks (Chen et al., 2023a; Puigcerver et al., 2023). More recently, the idea of conducting dynamic networks has been applied to image processing. Yu et al. (2021) proposed to select the appropriate route for each image region with a pathfinder, so that the input can be dynamically proceed. Zamfir et al. (2024) introduced the low-rank experts to formulate the inter-dependencies among the relevant feature projections for image super-resolution. Cao et al. (2023) proposed to extract effective and comprehensive information from the respective modalities of the RGB image and the infrared image with MoEs.

In this paper, we explore the MoEs from the perspective of perceptual cues integration for low-light image restoration, aiming at designing dedicated routers and experts to tackle couple-degraded image restoration.

3. Methodology

We propose to integrate perceptual cues with mixture-of-experts for low-light image restoration. In this section, we first present an overview of the proposed framework. Then, we describe dedicated MoEs with the perceptual cues in detail. Subsequently, we develop three model variants of the IPCMoE for low-light image restoration. Finally, we introduce the loss functions.

3.1. Overview of the framework

As illustrated in Fig. 1 (b), the overall workflow of the proposed IPCMoE framework adopts a progressive multi-stage architecture, in which the enhancement MoE (E-MoE) and texture memorial MoE (T-MoE) are executed in parallel to process multi-scale features. Concretely, as depicted in Fig. 2, given the input image I, we initially employ two 3×3 convolutions to extract low-level features. Subsequently, the extracted feature traverses through a series of enhancement stages for progressive processing. Specifically, given the input $\mathbf{f}_i^{\mathrm{in}}$ of the ith enhancement stage $\mathcal{S}_i$, a multi-scale feature extraction module (Li et al., 2024c; Liu et al., 2020) is firstly applied to encode the enriched feature $\mathbf{f}_i^{\mathrm{Ms}}$.

After that, the multi-scale feature is parallelly fed into the enhancement MoE (E-MoE) and the texture memorial MoE (T-MoE). To integrate the perceptual cues, we develop the auxiliary task of texture map prediction to encode the texture feature memories, and correspondingly propose the texture memorial MoE that assembles the historical memory $\mathbf{m}_{i-1}$ and the multi-scale feature for generating texture cues $\mathbf{c}_i^{\mathrm{Te}}$ as follows:

$$\mathbf{c}_i^{\mathrm{Te}} = \text{T-MoE}(\mathbf{f}_i^{\mathrm{Ms}}, \mathbf{m}_{i-1}). \tag{1}$$

Meanwhile, we build the auxiliary task of thumbnail enlightening to extract enlightening cues $\mathbf{c}_i^{\mathrm{En}}$ and inject them into the enhancement MoE to guide the expert selection process. We design the enhancement MoE (E-MoE) to provide adaptive feature processing with the integration of texture cues $\mathbf{c}_i^{\mathrm{Te}}$ and enlightening cues $\mathbf{c}_i^{\mathrm{En}}$. The output of the enhancement stage $\mathcal{S}_i$ can be obtained as follows:

$$\mathbf{f}_i^{\mathrm{out}} = \text{E-MoE}(\mathbf{f}_i^{\mathrm{Ms}}, \mathbf{c}_i^{\mathrm{En}}, \mathbf{c}_i^{\mathrm{Te}}). \tag{2}$$

Ultimately, the output from the last enhancement stage is processed with two 3×3 convolutions to generate the normal-light sharp image result $\hat{Y}$.

3.2. Texture memorial MoE (T-MoE)

Information cues for texture restoration are easily smeared in deep networks, leading to oversmooth results. As a result, a widely adopted approach for preserving detail is to concatenate shallow features to deep network layers. However, this manner raises the risk of introducing artifacts, as low-light and blur degradations may also be retained through the shallow features. To meet these challenges, we propose a texture memorial MoE (T-MoE) as follows.

Formulation of the T-MoE: Given the multi-scale feature $\mathbf{f}_i^{\mathrm{Ms}}$ as input, we design an extended router ($\mathcal{G}^{\mathrm{Te}}$) to select proper texture restoration experts ($\mathcal{E}^{\mathrm{Te}}$) for texture restoration. In the enhancement stage $\mathcal{S}_i$, the texture cues can be obtained as follows:

$$\mathbf{c}_i^{\mathrm{Te}} = \mathrm{MemUp}(\mathbf{f}_i^{\mathrm{Te}}), \tag{3}$$

where the $\mathbf{f}_i^{\mathrm{Te}}$ is calculated as:

$$\mathbf{f}_i^{\mathrm{Te}} = \sum_{j=1}^{n^t} \mathcal{G}_i^{\mathrm{Te}}(\mathbf{f}_i^{\mathrm{Ms}})_j \cdot \mathcal{E}_{i,j}^{\mathrm{Te}}(\mathbf{f}_i^{\mathrm{Ms}}) + \mathcal{G}_i^{\mathrm{Te}}(\mathbf{f}_i^{\mathrm{Ms}})_{(n^t+1)} \cdot \mathbf{m}_{i-1}, \tag{4}$$

where $\mathcal{G}_i^{\mathrm{Te}}(\cdot)$ is the extended router, $\mathcal{E}_{i,j}^{\mathrm{Te}}$ is the jth texture restoration expert, $\mathbf{m}_{i-1}$ is the texture memory, n^t is the number of texture restoration experts, and the $\mathrm{MemUp}(\cdot)$ is the texture memory update operation.

Our T-MoE takes k^t $(1 \leq k^t \leq n^t)$ texture restoration experts to process the multi-scale feature. Additionally, the texture memory $\mathbf{m}_{i-1}$ is adopted as the (n^t+1)th expert, which is always activated for calculation. In other words, the texture cues $\mathbf{c}_i^{\mathrm{Te}}$ is generated depending on $k^t + 1$ experts of the T-MoE. Specifically, in our extended router (Fig. 3), the multi-scale feature is spatially squeezed to 1×1. Subsequently, we employ an MLP layer to generate the indicator vector $\mathbf{h}^{\mathrm{Te}} \in \mathbb{R}^{1\times(n^t+1)}$ for

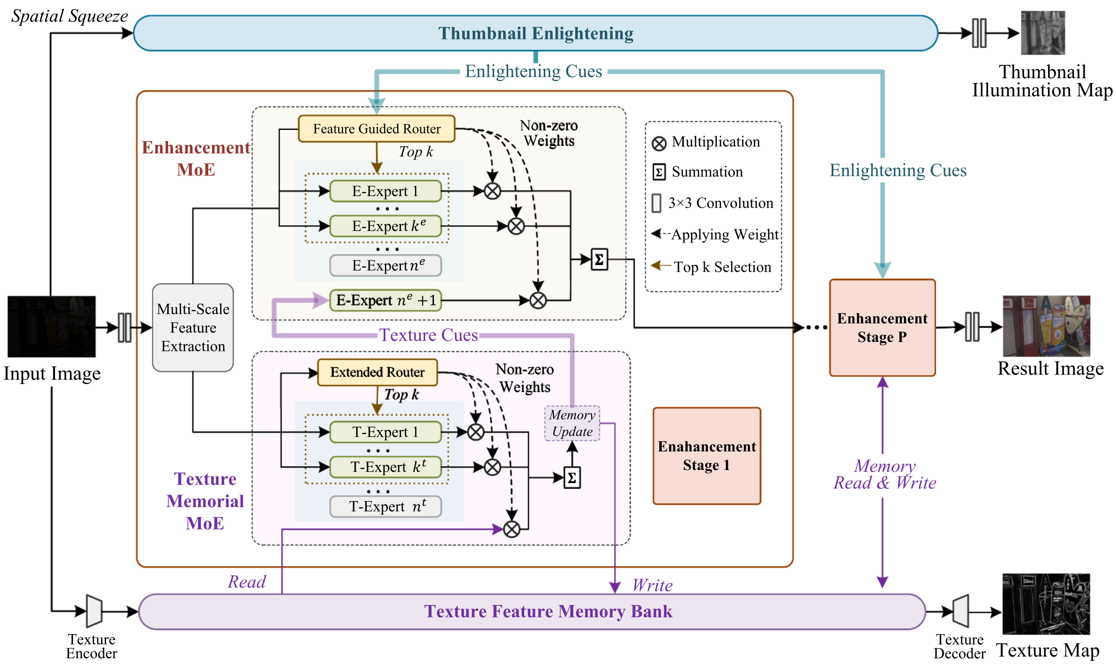


Fig. 2. Illustration of the architecture of the proposed method, which consists of a series of "enhancement stages". Each stage performs multi-scale feature extraction followed by parallel processing via the enhancement MoE and texture memorial MoE. In addition, a thumbnail enlightening task and a texture map prediction task are introduced to provide enlightening and texture cues, respectively. The enhancement MoE then adaptively refines features by integrating these perceptual cues.

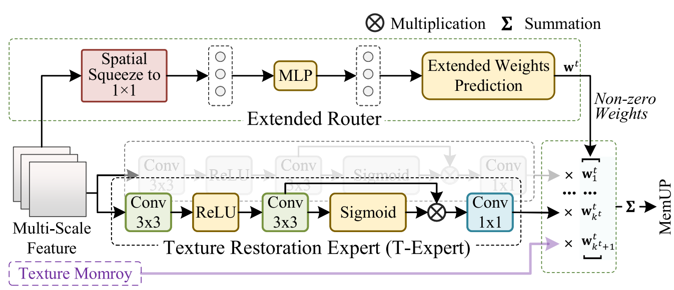


Fig. 3. Our texture memorial MoE (T-MoE) consists of an extended router and a series of texture restoration experts (T-Expert). Aggregated texture memory features are further treated as an additional expert, facilitating long-range modeling of multi-level representations across enhancement stages.

the extended weights prediction process (see Algorithm 1). As a result, the extended router $\mathcal{G}^{\mathrm{Te}}$ determines that k^t of n^t texture restoration experts are selected, and predicts the corresponding weights $\mathbf{w}^t \in \mathbb{R}^{1\times(n^t+1)}$ for the experts and the texture memory.

Texture Restoration Expert (T-Expert): To capture and transform degraded patterns, we design dedicated experts to manipulate features with a focus on spatial dimension (Fig. 3). Given the input feature $\mathbf{f}_i^{\mathrm{Ms}}$, a typical Conv-ReLU-Conv is conducted to transform features as follows:

$$\mathbf{f}_f^{\mathrm{Te}} = C_{3\times3}(\tau(C_{3\times3}(\mathbf{f}_i^{\mathrm{Ms}}))), \tag{5}$$

where $C_{3\times3}$ denotes the 3×3 convolution, and $\tau(\cdot)$ represents the ReLU function. Then, we employ a spatial attention mechanism in our T-Expert as follows:

$$\mathcal{E}_i^{\mathrm{Te}}(\mathbf{f}_f^{\mathrm{Te}}) = C_{1\times1}(\mathbf{f}_f^{\mathrm{Te}} \otimes \mathrm{Sigmoid}(\mathbf{f}_f^{\mathrm{Te}})), \tag{6}$$

where $C_{1\times1}$ denotes the 1×1 convolution, and $\otimes$ is the element-wise multiplication. In Eq. (6), the attention map $\mathrm{Sigmoid}(\mathbf{f}_f^{\mathrm{Te}})$ is obtained through a parameter-free activation function to focus on information-rich regions (Wan et al., 2024).

Texture Memory Update: To preserve the informative details in the image enhancement procedure, we devise the "texture feature memory bank". As shown in Fig. 2, the texture encoder adopts the low-quality image as input, and the texture feature memory bank is read and written by the T-MoE at every enhancement stage. Accordingly, the texture feature memory $\mathbf{m}_i$ is obtained as follows:

$$\mathbf{m}_i = \begin{cases} \mathbf{f}_i^{\mathrm{Te}} + \alpha_i \cdot \mathbf{m}_{i-1} & i \geq 1, \\ \mathrm{TE}(I) & i = 0, \end{cases} \tag{7}$$

where $\mathbf{f}_i^{\mathrm{Te}}$ is computed by Eq. (4), and the α_i is a learnable momentum value. We regard the texture feature memory $\mathbf{m}_i$ as the texture cues $\mathbf{c}_i^{\mathrm{Te}}$ at current enhancement stage. The initial memory $\mathbf{m}_0$ is obtained by the texture encoder $\mathrm{TE}(\cdot)$, consisting of two 3×3 convolutions followed by a ReLU function and a 1×1 convolution.

In our design, $\mathbf{m}_i$ is extracted from every enhancement stage for detail preservation. An inescapable question is: *how do we ensure that the extracted memories can encode the texture information?* To this end, we present the auxiliary task of texture map prediction to empower intermediate features for texture representation. The auxiliary task predicts the texture map as follows:

$$\hat{T} = \mathrm{TD}(\mathbf{m}_p), \tag{8}$$

where $\mathrm{TD}(\cdot)$ denotes our texture decoder, consisting of two 3×3 convolutions followed by a ReLU function and a 1×1 convolution, and $\hat{T}$ is the predicted texture map.

Algorithm 1: Extended weights prediction.

Input: Number of experts n, Top-k value k, Indicator vector $\mathbf{h} \in \mathbb{R}^{1\times(n+1)}$.
Output: Gating weights $\mathbf{w} = \{\mathbf{w}_1, \dots, \mathbf{w}_{(n+1)}\}$.

```
▷ Dense expert collaboration:
if k = n then
    w = Softmax(h); return w;
▷ Selective expert collaboration:
if k < n then
    Θ_ind ← TopKIndices(Softmax(h[1 : n]), k);   // Find topk indices among n experts.
    Θ_ind ← {n + 1};
    // Append the additional expert indice.
    w' = Softmax(h);
    foreach w'_i in w' do
        if i in Θ_ind then
            w_i = w'_i;                          // Keep k+1 weights.
        else
            w_i = 0;                             // Discarding others.
    return gating weights w;
```

3.3. Enhancement MoE (E-MoE)

A typical dilemma in image enhancement is that the visual degradation varies from one scene to another. Considering a couple-degraded low-light image, if the enlightening operation corresponding to a predefined illumination adjustment is applied to an arbitrary low-light input, there may appear under-enhanced or over-enhanced regions. Similarly, applying inappropriate restoration can introduce unintended artifacts, leading to poor-quality results. In addition, increasing image brightness while neglecting texture restoration may impair the detail restoration (Li et al., 2024c; Ye et al., 2024b).

Our motivation is to exploit the dynamic weighting mechanism of MoE to adaptively process the image. Therefore, we propose the enhancement MoE with the integration of perceptual cues, which allows an adaptive combination of the feature enlightening and valuable texture aggregation at different enhancement stages.

Formulation of the E-MoE: The proposed enhancement MoE takes the multi-scale feature $\mathbf{f}_i^{\mathrm{Ms}}$, texture cues feature $\mathbf{c}_i^{\mathrm{Te}}$ and enlightening cues feature $\mathbf{c}_i^{\mathrm{En}}$ as inputs. Our E-MoE can be formulated as follows:

$$\begin{aligned}\mathbf{f}_i^{\mathrm{En}} = &\sum_{j=1}^{n^e} \mathcal{G}_i^{\mathrm{En}}(\mathbf{f}_i^{\mathrm{Ms}}, \mathbf{c}_i^{\mathrm{En}})_j \cdot \mathcal{E}_{i,j}^{\mathrm{En}}(\mathbf{f}_i^{\mathrm{Ms}}) \\ &+ \mathcal{G}_i^{\mathrm{En}}(\mathbf{f}_i^{\mathrm{Ms}}, \mathbf{c}_i^{\mathrm{En}})_{(n^e+1)} \cdot \mathcal{E}_{i,(n^e+1)}^{\mathrm{En}}(\mathbf{c}_i^{\mathrm{Te}}),\end{aligned} \tag{9}$$

where $\mathcal{G}_i^{\mathrm{En}}(\cdot)$ is the feature guided router, $\mathcal{E}_{i,j}^{\mathrm{En}}$ is the jth enhancement expert, and n^e is the number of enhancement experts. $\mathbf{c}_i^{\mathrm{Te}}$ and $\mathbf{c}_i^{\mathrm{En}}$ denote the texture cues and enlightening cues, respectively.

Enlightening Cues Integration: Downsampled images can naturally reduce the perception of noise and blur degradations. This allows the model to decouple perception by focusing only on illumination patterns. We extract the enlightening cues by constructing an auxiliary task of thumbnail enlightening. Specifically, the low-quality image is spatially squeezed to 64×64 to obtain a thumbnail input. Subsequently, three 3×3 convolutions with a ReLU function are used for feature projection, and two 3×3 convolutions are employed to estimate the thumbnail illumination map. In this process, the projected intermediate features are extracted as enlightening cues $\mathbf{c}_i^{\mathrm{En}}$.

To integrate enlightening cues, we propose a feature guided router ($\mathcal{G}^{\mathrm{En}}$) for dynamic expert selection. As shown in Fig. 4, the enlightening cues $\mathbf{c}_i^{\mathrm{En}}$ are processed with a 3×3 convolution, and the multi-scale feature $\mathbf{f}_i^{\mathrm{Ms}}$ is spatially squeezed to 64×64 to match the size of $\mathbf{c}_i^{\mathrm{En}}$ for concatenation. The concatenated feature is then processed by a Conv-ReLU-Conv (CRC) block and spatially squeezed to 1×1. Finally, a multi-layer perceptron (MLP) is employed to generate the indicator vector $\mathbf{h}^{\mathrm{En}} \in \mathbb{R}^{1\times(n^e+1)}$ for the extended weights prediction process (see Algorithm 1).

Texture Cues Integration: As shown in Fig. 4, we propose to integrate the texture cues with the enhancement MoE. Specifically, the multi-scale feature is processed by the selected k^e ($1 \le k^e \le n^e$) enhancement experts, and the texture cues feature is handled by an extra expert that is always involved in the calculation. To this end, we reuse our extended weights prediction process (Algorithm 1) to select k^e of n^e enhancement experts and output the corresponding gating weights $\mathbf{w}^e \in \mathbb{R}^{1\times(n^e+1)}$. As a result, the enhanced feature and texture cues feature can be adaptively fused by the predicted gating weights based on each individual input.

Enhancement Expert (E-Expert): Given the multi-scale feature $\mathbf{f}_i^{\mathrm{Ms}}$ as input, we build a simple conv-based enhancement expert as follows:

$$\mathcal{E}_{i,j}^{\mathrm{En}}(\mathbf{f}_i^{\mathrm{Ms}}) = C_{3\times3}(\tau(C_{1\times1}(\tau(C_{3\times3}(\mathbf{f}_i^{\mathrm{Ms}}))))), \tag{10}$$

where $C_{1\times1}$ and $C_{3\times3}$ denote the 1×1 and 3×3 convolution, respectively. The $\tau(\cdot)$ represents the ReLU function.

3.4. Build models for low-light image restoration

In this paper, we focus not only on low-light enhancement (LLE), but also on couple-degraded scenarios, i.e., joint low-light enhancement and deblurring (LLE-Deblur), and joint low-light enhancement super-resolution (LLE-SR). We develop the following model variants.

IPCMoE-L for LLE: Our IPCMoE-L is built with $p = 2$ enhancement stages for the low-light image enhancement task. $k^t = 2$ of $n^t = 6$ texture restoration experts are selected for collaboration in the T-MoE, and $k^e = 2$ of $n^e = 6$ enhancement experts are selected for collaboration in the E-MoE.

IPCMoE-D for LLE-Deblur: Our IPCMoE-D is built with $p = 4$ enhancement stages for joint low-light enhancement and deblurring. Similar to the IPCMoE-L, $k^t = 2$ of $n^t = 6$ texture restoration experts are selected for collaboration in the T-MoE, and $k^e = 2$ of $n^e = 6$ enhancement experts are selected for collaboration in the E-MoE.

IPCMoE-S for LLE-SR: Our IPCMoE-S is built with $p = 4$ enhancement stages for joint low-light enhancement and super-resolution. Similarly, we select $k^t = 2$ of $n^t = 6$ texture restoration experts for collaboration in the T-MoE, and $k^e = 2$ of $n^e = 6$ enhancement experts for collaboration in the E-MoE. For super-resolution, we employ a learnable upsampling module from (Zhang et al., 2018) to empower our method for spatial upsampling.

3.5. Loss functions

Restoration Loss: Following the previous methods (Cho et al., 2021; Kong et al., 2023; Zhou et al., 2022), we adopt the L1 loss and perceptual loss to supervise the low-light image restoration as follows:

$$\mathcal{L}_{\mathrm{restore}} = ||\hat{Y} - Y||_1 + \lambda_{\mathrm{perc}} \cdot ||\Phi(\hat{Y}) - \Phi(Y)||_1, \tag{11}$$

where $\hat{Y}$ is the result produced by our method, and Y denotes the reference image. $\Phi(\cdot)$ extracts multi-scale features with a pretrained VGG-19 network for perceptual loss calculation, and $\lambda_{\mathrm{perc}} = 0.01$ is the empirical setting.

Auxiliary Task Loss: The following loss is developed to build the auxiliary tasks to extract perceptual cues:

$$\mathcal{L}_{\mathrm{aux}} = \mathcal{L}_{\mathrm{thumb}} + \mathcal{L}_{\mathrm{texture}}, \tag{12}$$

where the $\mathcal{L}_{\mathrm{thumb}}$ constraints the auxiliary task of thumbnail enlightening, and the $\mathcal{L}_{\mathrm{texture}}$ is designed for texture map prediction. In detail, the $\mathcal{L}_{\mathrm{thumb}}$ is as follows:

$$\mathcal{L}_{\mathrm{thumb}} = ||\hat{Y}_{\mathrm{thumb}} - \mathcal{V}(Y_{\downarrow})||_1, \tag{13}$$

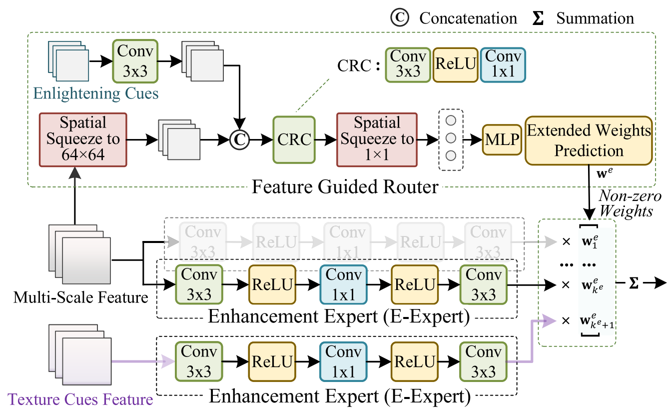


Fig. 4. Our enhancement MoE (E-MoE) consists of a feature guided router and a series of enhancement experts (E-Expert). The feature guided router integrates the enlightening cues for expert selection and weights prediction. The texture cues feature is processed with an extra expert for dynamical integration.

Table 1
Evaluation on the SID, SDSD-indoor, and SDSD-outdoor datasets. The **<u>bold-underlined</u>**/**bold**/<u>underlined</u> fonts indicate best three performance.

Type	Methods	SID			SDSD-indoor			SDSD-outdoor		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
LLE	SNR-Net (Xu et al., 2022) *CVPR'22*	21.348	0.550	0.434	26.135	0.815	0.198	19.217	0.657	0.225
	UHDFour (Li et al., 2023) *ICLR'23*	<u>22.744</u>	0.651	<u>0.333</u>	**29.133**	<u>0.895</u>	**0.096**	23.283	0.771	0.169
	FourLLIE (Wang et al., 2023) *ACM MM'23*	18.424	0.513	0.476	24.736	0.826	0.146	24.674	0.787	<u>0.158</u>
	LLFormer (Wang et al., 2023b) *AAAI'23*	**22.830**	<u>0.656</u>	0.342	26.284	0.825	0.232	**29.180**	0.841	**0.137**
	RetinexMamba (Bai et al., 2024) *ICONIP'24*	22.449	<u>0.656</u>	0.341	28.443	0.894	0.119	<u>28.521</u>	<u>0.859</u>	0.180
	BiFormer (Xu et al., 2024) *IEEE TMM'24*	22.490	0.638	0.368	<u>28.730</u>	**0.899**	<u>0.100</u>	26.132	0.841	0.180
Unified	Restormer (Zamir et al., 2022) *CVPR'22*	22.017	0.645	0.350	28.494	0.892	0.124	27.992	**<u>0.868</u>**	0.189
	Uformer (Wang et al., 2022) *CVPR'22*	18.546	0.577	–	23.172	0.859	–	23.850	0.748	–
	MambaIR (Guo et al., 2024) *ECCV'24*	22.023	**0.658**	**0.326**	25.135	0.876	0.130	27.527	0.851	0.177
	IPCMoE-L (Ours)	**<u>23.156</u>**	**<u>0.661</u>**	**<u>0.271</u>**	**<u>29.241</u>**	**<u>0.903</u>**	**<u>0.082</u>**	**<u>29.290</u>**	**<u>0.868</u>**	**<u>0.114</u>**

where $\hat{Y}_{\text{thumb}}$ is the predicted thumbnail illumination map, and $Y_{\downarrow}$ is the 64×64 spatially squeezed reference image. $\mathcal{V}(\cdot)$ extracts the illumination channel in HSV color space.

The loss function $\mathcal{L}_{\text{texture}}$ provides an indirect constraint on extracting texture memory features:

$$\mathcal{L}_{\text{texture}} = ||\hat{T} - \text{Edge}(Y)||_2^2, \tag{14}$$

where $\hat{T}$ is the predicted texture map, and $\text{Edge}(\cdot)$ extracts texture edges of the reference image Y with Sobel operator.

Total Loss: The total loss $\mathcal{L}_{\text{total}}$ for optimization is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{restore}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{aux}}, \tag{15}$$

where $\lambda_{\text{aux}} = 0.15$ is the trade-off hyperparameter.

4. Experiments

In this section, we first present the necessary technical details for implementation. Subsequently, we perform the qualitative and quantitative evaluations on benchmarks for low-light enhancement, joint low-light enhancement and deblurring, as well as joint low-light enhancement and super-resolution. Finally, ablation experiments are conducted to validate the effectiveness of our method.

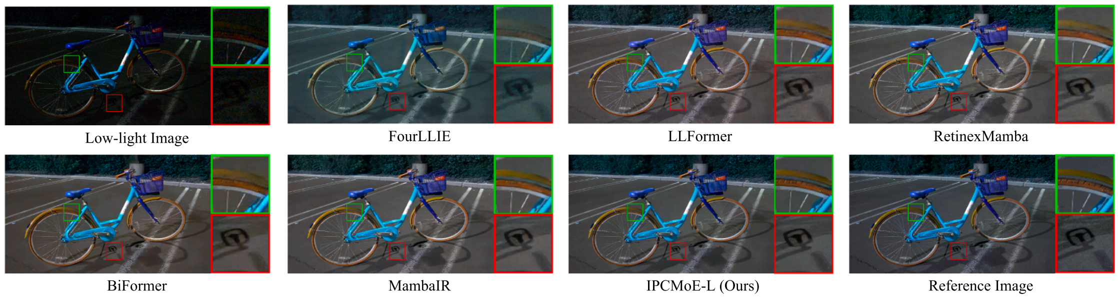
4.1. Implementation details

Multi-scale Feature Extraction: we extract the multi-scale feature with an off-the-shelf multi-scale feature extraction (MSFE) module (Li et al., 2024c). Given the input feature, the MSFE module starts with a 3×3 convolution, then three network branches process the features in $\times1$, $\times2$ downscaled, and $\times4$ downsampled sizes, respectively. Subsequently, each scaled feature is handled with three ResGroups, consisting of five ResBlocks with a residual connection. Each ResBlock consists of a 3×3 convolution, a PReLU function, and a 3×3 convolution with residual connection. After that, the processed features in $\times2$ and $\times4$ downsampled get upscaled and concatenated followed by a 3×3 convolution, and later concatenated with the processed $\times1$ features and the input feature. Finally, 3×3 and 1×1 convolutions are adopted for channel reduction. The channel dimension is set to 48, and bilinear interpolations are used for feature downsampling and upsampling.

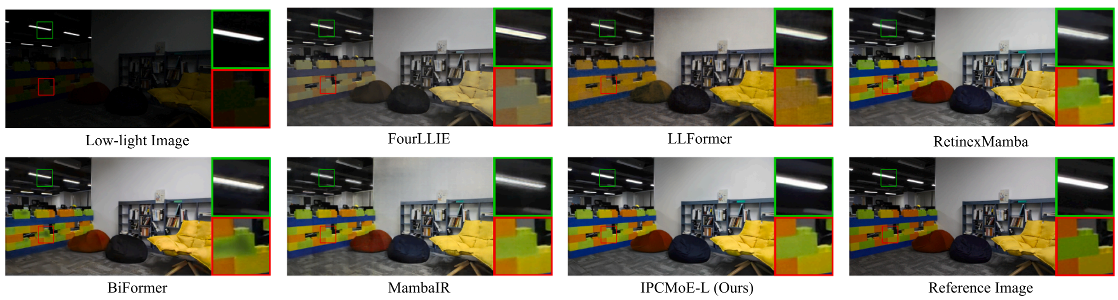
Training Details: We implement our model using PyTorch 2.0 on a PC equipped with a single NVIDIA A100 GPU. We use Adam as the optimizer for 480K iterations with an initial learning rate 10^{-4} gradually reduced to 10^{-7} with the cosine annealing. We employ the progressive training strategy (Zamir et al., 2022) by gradually increasing the patch size and reducing the batch size to $[(128^2, 12), (160^2, 10), (192^2, 8), (256^2, 6), (320^2, 4), (384^2, 2)]$ at every $80K$ iterations. In addition, common data augmentation techniques (i.e., horizontal flipping and rotation) are randomly applied to the training data.

4.2. Comparison on low-light image enhancement (LLE)

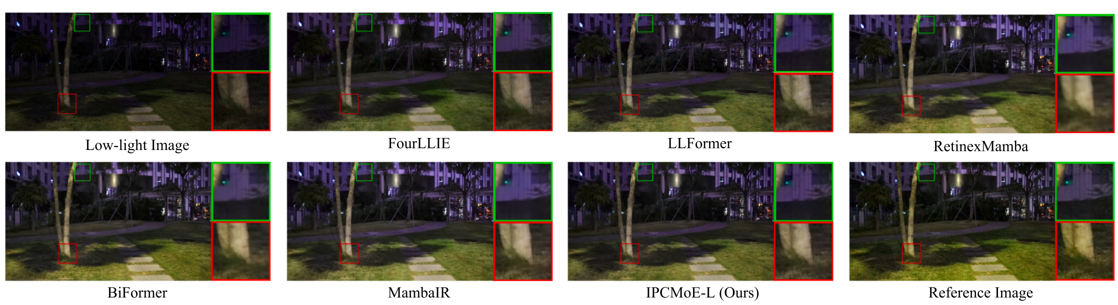
We compare our IPCMoE-L with the SOTA methods for LLE task. Specifically, the following low-light enhancement (LLE) methods are compared: SNR-Net (Xu et al., 2022), UHDFour (Li et al., 2023), FourLLIE (Wang et al., 2023), LLFormer (Wang et al., 2023b), RetinexMamba (Bai et al., 2024), and BiFormer (Xu et al., 2024). In addition, two unified image processing methods are also adopted for comparison: Restormer (Zamir et al., 2022), and MambaIR (Guo et al., 2024). For evaluation, we utilize three reference-based metrics: PSNR, SSIM, and LPIPS, and the quantitative results are shown in Table 1.



(a) Visual comparison on the SID dataset.



(b) Visual comparison on the SDSD-indoor dataset.

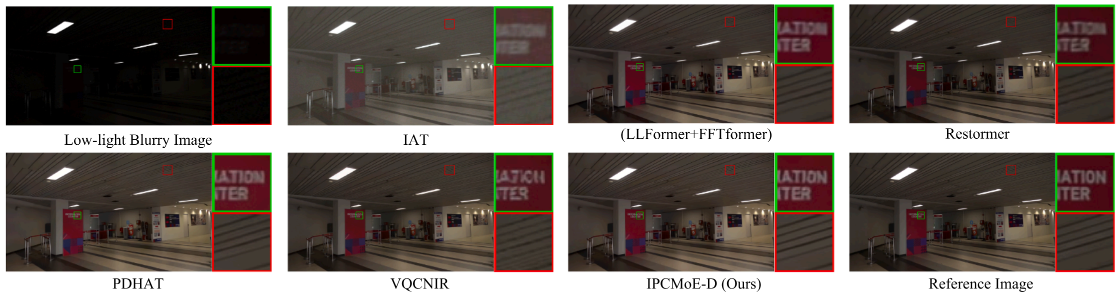


(c) Visual comparison on the SDSD-outdoor dataset.

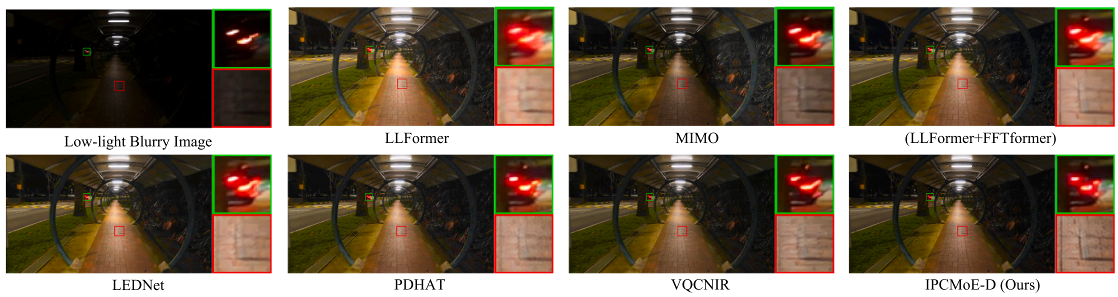
Fig. 5. Visual comparison on the low-light enhancement task. The zoomed-in regions are shown right side of each result image.

Table 2
Evaluation on the LOL-Blur and the Real-LOL-Blur datasets. For the LLE methods, the deblurring methods, the jointly retrained methods and the unified methods, the best results are marked in underlined. For the LLE-Deblur methods, the **bold-underlined**/**bold**/underlined fonts indicate best three performance.

Type	Methods	LOL-Blur			Real-LOL-Blur		
		PSNR↑	SSIM↑	LPIPS↓	NIQE↓	BRISQUE↓	PI↓
LLE	IAT (Cui et al., 2022) *BMVC'22*	21.404	0.728	0.424	5.711	26.105	5.462
	FourLLIE (Wang et al., 2023) *ACM MM'23*	19.808	0.699	0.439	5.484	35.815	5.749
	LLFormer (Wang et al., 2023b) *AAAI'23*	24.549	0.798	0.354	6.107	39.803	5.998
Deblur	MIMO (Cho et al., 2021) *ICCV'21*	22.618	0.827	0.269	5.871	43.069	5.539
	SharpFormer (Yan et al., 2023) *IEEE TIP'23*	23.997	0.871	0.233	4.865	37.330	4.645
	FFTformer (Kong et al., 2023) *CVPR'23*	24.381	0.867	0.238	5.173	41.129	4.991
LLE&Deblur	(LLFormer (Wang et al., 2023b) + FFTformer (Kong et al., 2023))	24.998	0.854	0.224	6.501	48.924	5.567
Unified	NAFNet (Chen et al., 2022a) *ECCV'22*	23.338	0.836	0.316	5.325	38.141	5.186
	Restormer (Zamir et al., 2022) *CVPR'22*	26.379	0.868	0.250	5.731	39.373	5.304
	MIRNet-V2 (Zamir et al., 2023) *IEEE TPAMI'23*	25.654	0.840	0.295	7.341	53.189	6.451
LLE-Deblur	LEDNet (Zhou et al., 2022) *ECCV'22*	26.059	0.852	0.217	5.337	46.022	4.881
	Ye *et. al* (Ye et al., 2024b) *ESWA'24*	26.730	0.866	0.199	4.442	-	4.430
	SSFlow (Li et al., 2024) *CSSP'24*	22.360	0.706	0.529	6.759	37.982	6.080
	PDHAT (Li et al., 2024c) *IEEE TMM'24*	26.713	**0.885**	0.175	**4.203**	**25.995**	**3.724**
	VQCNIR (Zou et al., 2024) *AAAI'24*	**27.793**	0.875	**0.172**	4.599	35.916	3.806
	LIEDNet (Liu et al., 2025) *IEEE TCSVT'25*	27.425	0.882	0.184	5.129	39.739	4.801
	IPCMoE-D (Ours)	**28.273**	**0.900**	**0.164**	**4.194**	**23.366**	**3.641**



(a) Visual comparison on the synthetic LOL-Blur dataset.



(b) Visual comparison on the real-captured Real-LOL-Blur dataset.

Fig. 6. Visual comparison on the joint low-light enhancement and deblurring task. The zoomed-in regions are shown right side of each result image.

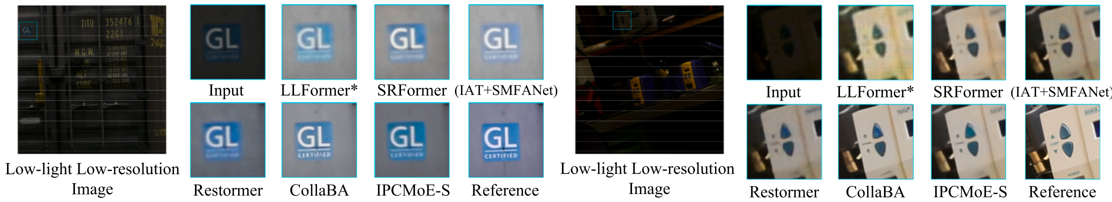
Table 3
Evaluation on the RELLISUR ×2 and the RELLISUR ×4 datasets. "*" denotes the compared method that is trained combining with an upsampling module (Zhang et al., 2018). For the LLE methods, the super-resolution methods, the jointly retrained methods and the unified methods, the best results are marked in <u>underlined</u>. For the LLE-SR methods, the <u>**bold-underlined**</u>/**bold**/<u>underlined</u> fonts indicate best three performance.

Type	Methods	RELLISUR ×2			RELLISUR ×4		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
LLE	IAT* (Cui et al., 2022) ***BMVC'22***	19.748	0.717	0.575	15.870	0.679	0.763
	LLFormer* (Wang et al., 2023b) ***AAAI'23***	<u>20.385</u>	<u>0.738</u>	<u>0.513</u>	<u>19.471</u>	<u>0.752</u>	<u>0.632</u>
SR	SRFormer (Zhou et al., 2023) ***ICCV'23***	19.110	0.745	0.444	18.252	0.746	0.595
	HAT (Chen et al., 2023b) ***CVPR'23***	<u>21.141</u>	<u>0.761</u>	<u>0.412</u>	18.103	0.741	0.625
	SMFANet (Zheng et al., 2024) ***ECCV'24***	20.164	0.748	0.436	<u>19.743</u>	<u>0.758</u>	<u>0.550</u>
LLE&SR	(IAT (Cui et al., 2022) + SMFANet (Zheng et al., 2024))	<u>20.470</u>	<u>0.754</u>	<u>0.437</u>	<u>19.283</u>	<u>0.751</u>	<u>0.573</u>
Unified	Restormer (Zamir et al., 2022) ***CVPR'22***	<u>21.893</u>	<u>0.768</u>	<u>0.388</u>	<u>20.919</u>	<u>0.755</u>	<u>0.503</u>
	MIRNet-V2 (Zamir et al., 2023) ***IEEE TPAMI'23***	17.913	0.702	0.507	17.234	0.717	0.603
LLE-SR	LUCN (Cheng et al., 2022) ***IEEE TCSVT'22***	–	–	–	19.824	0.652	0.708
	CollaBA (Gao et al., 2024) ***PR'24***	<u>21.621</u>	<u>**0.787**</u>	<u>**0.247**</u>	<u>20.423</u>	0.734	<u>**0.371**</u>
	MSIRNet (Ye et al., 2024a) ***Info. Fusion'24***	21.330	<u>0.749</u>	0.365	20.320	**0.735**	<u>0.437</u>
	UltraIS (Gao et al., 2025) ***IEEE TNNLS'25***	**22.264**	0.743	**0.254**	**21.036**	0.726	<u>**0.371**</u>
	IPCMoE-S (Ours)	<u>**22.282**</u>	**0.782**	<u>0.334</u>	<u>**21.157**</u>	<u>**0.776**</u>	0.467

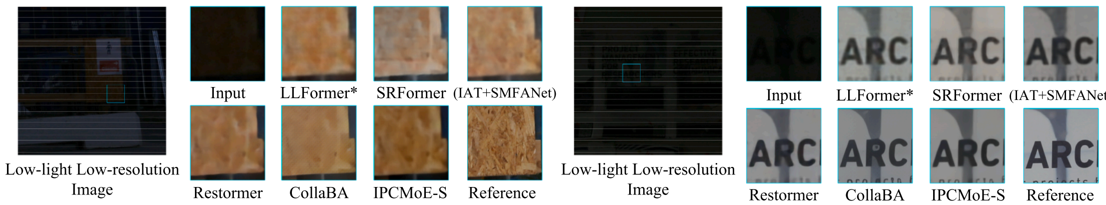
Comparison on SID (Chen et al., 2018) Dataset: The SID datasets feature short- and long-exposure images with significant noise. The SID subset consists of 2697 paired shortand long-exposure images captured using the Sony α7S II camera. This dataset is partitioned into 2099 pairs for training and the other 598 pairs for testing.

As can be seen in Table 1, our IPCMoE-L demonstrates superior performance compared to LLE models on SID, outperforming RetinexMamba by 0.707 dB in PSNR. Similarly, compared to the Restormer, an unified model, our model outperforms it by 1.139 dB in PSNR and 0.016 in SSIM. As shown in Fig. 5 (a), the result of FourLLIE exhibits a color shift, while LLFormer and RetinexMamba smear the details of the bicycle tire. Conversely, our approach preserves fine details while preventing excessive smoothing.

Comparison on SDSD (Wang et al., 2021) Dataset: This dataset is captured with a Canon EOS 6D Mark II camera equipped with an ND filter to regulate light exposure. We follow the (Niu et al., 2025) to introduce a static version of the SDSD dataset, which comprises both indoor and outdoor subsets, characterized by extremely low-brightness conditions. Specifically, the SDSD-indoor is partitioned into 1655 pairs for training and the 308 pairs for testing; the SDSD-outdoor is partitioned into 2650 pairs for training and the 500 pairs for testing. As can be seen in Table 1, our IPCMoE-L demonstrates superior performance compared to LLE models and unified models on SDSD, achieving six best performance values out of all metrics. To provide a more intuitive comparison, we showcase a challenging indoor scene with light sources in Fig. 5 (b). As can be seen, BiFormer produces unintended artifacts, MambaIR fails to handle light source regions properly, and our method produces a high-quality result. In addition, a comparison of a nighttime outdoor scene is shown in Fig. 5 (c), where our IPCMoE-L preserves more details.



(a) Visual comparison on the RELLISUR ×2 dataset.



(b) Visual comparison on the RELLISUR ×4 dataset.

Fig. 7. Visual comparison on the joint low-light enhancement and super-resolution task. The zoomed-in regions are shown right side of the input image.

4.3. Comparison on joint low-light enhancement and deblurring (LLE-Deblur)

We compare our IPCMoE-D with the SOTA methods, including three low-light enhancement (LLE) methods: IAT (Cui et al., 2022), FourLLIE (Wang et al., 2023), and LLFormer (Wang et al., 2023b); three image deblurring (Deblur) methods: MIMO (Cho et al., 2021), SharpFormer (Yan et al., 2023), and FFTformer (Kong et al., 2023); three unified image processing methods: NAFNet (Chen et al., 2022a), Restormer (Zamir et al., 2022), and MIRNet-V2 (Zamir et al., 2023). To investigate the efficiency of the joint training strategy, we combine the LLFormer (Wang et al., 2023b) and FFTformer (Kong et al., 2023) to build the "(LLFormer + FFTformer)". To this end, all these methods are *retrained* on the LOL-Blur training set using publicly available source code. In addition, five SOTA methods specializing in joint low-light enhancement and deblurring (LLE-Deblur) are adopted for comparison: LEDNet (Ye et al., 2024b; Zhou et al., 2022), SSFlow (Li et al., 2024), PDHAT (Li et al., 2024c), and VQCNIR (Zou et al., 2024).

Comparison on LOL-Blur Dataset: All compared models are trained on the synthetic dataset LOL-Blur, consisting of 10,200 image pairs for training and 1800 image pairs for testing. For paired data evaluation, we utilize three reference-based metrics: PSNR, SSIM, and LPIPS. As can be seen in Table 2, the proposed IPCMoE-D demonstrates superior performance over recent state-of-the-art LLE-Deblur methods PDHAT (Li et al., 2024c) and VQCNIR (Zou et al., 2024). We also provide a visible comparison in Fig. 6 (a).

Comparison on Real-LOL-Blur Dataset: This dataset consists of 1354 real-world night blurry images and no reference images are available. Therefore, all compared models are trained on LOL-Blur training set and tested on the Real-LOL-Blur. We employ three widely-used no-reference image quality assessment (IQA) methods for evaluation: NIQE, BRISQUE, and PI (Blau et al., 2018). The lower values of these three metrics indicate higher image quality. As can be seen in Table 2, our IPCMoE-D achieves the best performance on NIQE, BRISQUE, and PI metrics. The qualitative results of real-world performance are shown in Fig. 6. As can be observed, LLFormer and LEDNet tend to amplify light sources, resulting in halo artifacts in the output images, while VQCNIR fails to eliminate the blurry of the taillights.

4.4. Comparison on joint low-light enhancement and super-resolution (LLE-SR)

We compare our IPCMoE-S with the SOTA methods for LLE-SR task. Specifically, the following methods are introduced for comparison, two low-light enhancement (LLE) methods: IAT (Cui et al., 2022) and LLFormer (Wang et al., 2023b); In addition, three super-resolution methods: SRFormer (Zhou et al., 2023), HAT (Chen et al., 2023b), and SMFANet (Zheng et al., 2024); unified image processing methods: Restormer (Zamir et al., 2022), and MIRNet-V2 (Zamir et al., 2023). To investigate the efficiency of the joint training strategy, we combine the IAT (Wang et al., 2023b) and SMFANet (Zheng et al., 2024) to build the "(IAT + SMFANet)". To this end, all these methods are *retrained* on the RELLISUR (Aakerberg et al., 2021) training set using publicly available source code. In addition, four SOTA methods specializing in joint low-light enhancement and super-resolution (LLE-SR) are adopted for comparison: LUCN (Cheng et al., 2022), DiD (Gao et al., 2023), CollaBA (Gao et al., 2024), MSIRNet (Ye et al., 2024a). For evaluation, we utilize three reference-based metrics: PSNR, SSIM, and LPIPS.

We obtain the RELLISUR (Aakerberg et al., 2021) dataset for evaluation. This dataset is a collection of image sequences containing real ×1, ×2, and ×4 normal-light images, together with five real low-light images. All images are collected with a Canon EOS 6D camera. For the LLE-SR task, we adopt RELLISUR ×2 and RELLISUR ×4 subsets for comparison.

Comparison on RELLISUR ×2 Dataset: We follow the official option to adopt 3610 image pairs for training and 425 image pairs for testing, respectively. As can be seen from Table 3, our IPCMoE-S demonstrates superior performance compared to LLE models and, the unified models. Compared to the LLE-SR models, our method achieves the best PSNR value, and the second-best SSIM value. We provide a visible comparison in Fig. 7 (a), showing that the proposed method provides the high-fidelity restoration results. In the examples from the second case, it is evident that IPCMoE-S restores the original color and texture details without over-smoothing. In contrast, other methods exhibit varying degrees of insufficient details, with some color distortions as well.

Comparison on RELLISUR ×4 Dataset: Similar to the setting of RELLISUR ×2, there are 3610 image pairs used for training and 425 image pairs adopted for testing. The results shown in Table 3 demonstrate that our method achieves the best PSNR and SSIM values for joint low-light enhancement and super-resolution at scale of ×4. The visual comparison is shown in Fig. 7 (b). As can be seen, our method can preserve more valuable texture details in the LLE-SR task.

4.5. Ablation study

To validate the effectiveness of our approach, we conduct ablation experiments on the proposed mechanism, the designed modules and the loss function.

Architecture and Modules: As shown in Table 4 (1), we first investigate the performance of our baseline model, i.e., performing multi-scale feature extraction at each enhancement stage without involving

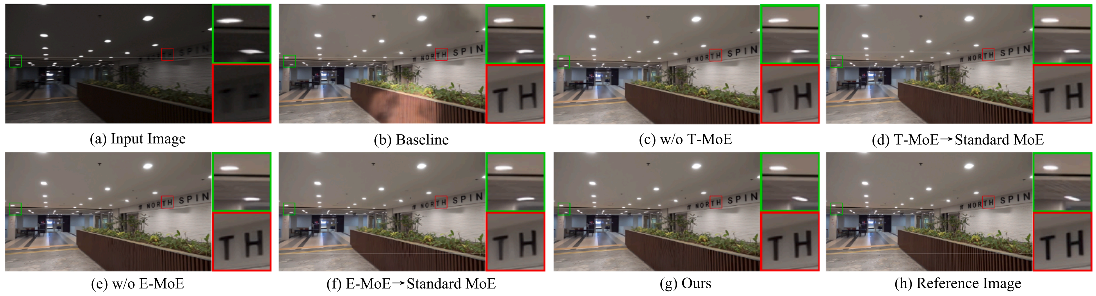


Fig. 8. Visual comparisons of the ablation study on network architecture and components. The zoomed-in regions (in green and red boxes) are shown right side of each result image.

Table 4
The ablation study of network components on the LOL-Blur dataset. PSNR, SSIM, and LPIPS are adopted for evaluation.

#	Model Variants	PSNR↑	SSIM↑	LPIPS↓
(1)	Baseline	24.127	0.868	0.186
(2)	w/o T-MoE	26.865	0.881	0.180
(3)	w/o Memory Usage in T-MoE	26.985	0.883	0.178
(4)	T-MoE→Standard MoE	27.402	0.899	0.165
(5)	w/o α_i in Eq. (7)	27.902	0.897	0.168
(6)	w/o E-MoE	26.683	0.884	0.176
(7)	E-MoE →Standard MoE	27.382	0.891	0.167
	Our Method	28.273	0.900	0.164

Table 5
The ablation study of loss functions on the LOL-Blur dataset. PSNR, SSIM, and LPIPS are adopted for evaluation.

#	Loss for Optimization	PSNR↑	SSIM↑	LPIPS↓
(1)	$\mathcal{L}_{\text{restore}}$	26.959	0.886	0.178
(2)	$\mathcal{L}_{\text{restore}} + \mathcal{L}_{\text{thumb}}$	27.560	0.887	0.179
(3)	$\mathcal{L}_{\text{restore}} + \mathcal{L}_{\text{texture}}$	27.220	0.894	0.167
(4)	$\mathcal{L}_{\text{restore}} + \mathcal{L}_{\text{thumb}} + \mathcal{L}_{\text{texture}}$	28.273	0.900	0.164

Table 6
Experimental analysis of adopting different expert collaboration mechanisms on LOL-Blur dataset.

#	Experts Collaboration	PSNR↑	SSIM↑	LPIPS↓
(1)	Dense	27.398	0.893	0.168
(2)	Random	26.905	0.886	0.176
(3)	Selective (Ours)	28.273	0.900	0.164

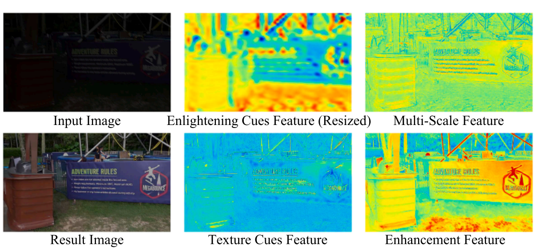


Fig. 9. Visualization of the intermediate features of our method based on a concrete case.

any MoE modules. This model provides a baseline performance of PSNR at 24.127. Subsequently, we set up three experiments to validate the effectiveness of the texture memorial MoE (T-MoE) in Table 4 (2)-(5). In Table 4 (2), our T-MoE is discarded, leading to lower PSNR and SSIM values. In Table 4 (3), we replace our T-MoE with a standard MoE (Shazeer et al., 2017). That is, the memory-related mechanisms are discarded, and the texture cues are generated by processing the multi-scale feature with a standard MoE. As a result, the PSNR value decreases from 28.273 to 26.985. Additionally, in our T-MoE, the texture memory acts as an additional expert, and is dynamically fused with other expert features to be flexibly retaining valuable information. To verify the effectiveness of this mechanism, we replace our extended router with a standard router, and fuse the texture memory in the T-MoE by adding it directly (Table 4 (4)). In addition, discarding the learnable memory momentum impairs texture restoration, as reflected by the decrease in the SSIM value (Table 4 (5)).

Furthermore, to validate the effectiveness of our enhancement MoE (E-MoE), two experiments are set up in Table 4 (6)-(7). Concretely, the E-MoE is removed in Table 4 (6), and the texture cues feature is accordingly added with the multi-scale feature in every enhancement stage. As a result, the PSNR decreases from 28.273 to 26.683, validating the effectiveness of our E-MoE. In Table 4 (7), the E-MoE is replaced by a standard MoE (Shazeer et al., 2017), this removes the feature guided router and adds texture cues directly to the output of the replaced standard MoE. The visible results of ablation are shown in Fig. 8.

Perceptual Cues Integration: In our method, we obtain the perceptual cues feature by constructing auxiliary tasks that aim to assist E-MoE in processing low-light low-quality images. As shown in Table 5, we discard the auxiliary losses for ablation. Specifically, Table 5 (1) takes $\mathcal{L}_{\text{restore}}$ for supervision, i.e., none of the auxiliary loss is involved in constraining the intermediate features. Table 5 (2) demonstrates adding the thumbnail enlightening task can benefit the expert selection process in the E-MoE, providing higher indicators. Table 5 (3) indicates that constraining the memory feature with the texture map prediction task leads to higher SSIM. In addition, we visualize the perceptual cues feature in Fig. 9 to provide an intuitive understanding. As can be seen, the enlightening cues feature describes the rough pattern of the illumination map, which is adopted to guide the expert selection in our approach. Furthermore, texture cue features encode valuable texture information that is integrated into the E-MoE. As a result, the processed enhancement feature can provide rich patterns and sufficient details.

5. Discussions

In this Section, we provide extended discussions on expert collaboration, model complexity and efficiency, extremely challenging cases, practical challenges, and future works.

5.1. Discussion on expert collaboration strategy

We explore the effects of employing different modes of expert cooperation for LLE-Deblur. In Table 6 (1), we use a non-sparse gating mechanism that employs all experts in calculating the output. As a result, n^t experts of the T-MoE and n^e experts of the E-MoE are fully and contribute to the output. It can be seen that, forcing the use of all experts weakens their proficiency in representing and handling the degradation. In Table 6 (2), $k^t = 2$ of $n^t = 6$ experts of the T-MoE and $k^e = 2$ of $n^e = 6$

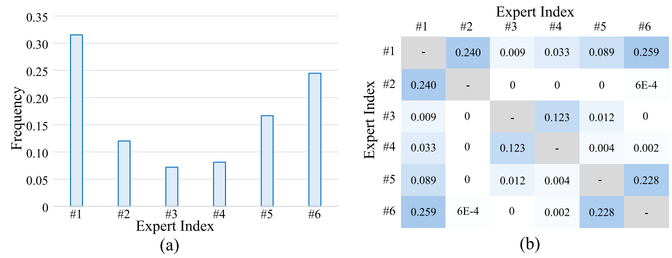


Fig. 10. Analysis of the expert selection and collaboration of our E-MoE. (a) is the frequency of expert selection. (b) is the matrix showing the probability of expert collaboration.

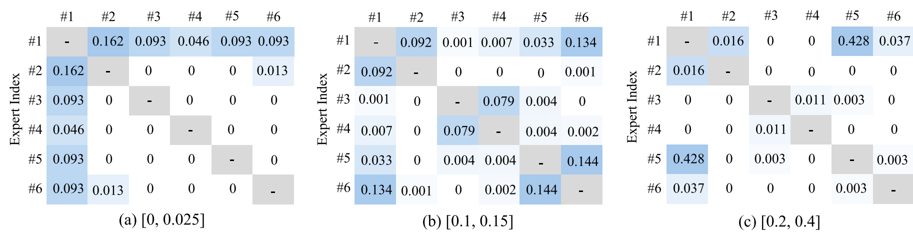


Fig. 11. Probability matrix of expert collaboration under different global illumination intensities.

of the E-MoE are *randomly* selected to sparsely process each input. Without using the topk selection mechanism, the random selection strategy weakens the variability between experts, reducing the flexibility of the MoEs to deal with hard scenarios.

5.2. Discussion on the expert collaboration from a perceptual perspective

The decision-making results of the feature guided router of the E-MoE are illustrated in Fig. 10. As can be observed, Fig. 10 (a) showcases a diverse selection of enhancement experts. Notably, enhancement experts #1, #5, and #6 have a higher frequency to be selected, as they may encode more generic features. Fig. 10 (b) illustrates the probability of expert collaboration, showing that expert #1 has a high generality and can be combined with all other experts. Experts #2-#6 are selective in collaboration, i.e., they collaborate with some experts and not with some others.

To unveil the capability of the proposed method in adaptively handling degradations, we show the probability of expert collaboration of the enhancement MoE (E-MoE) when processing images with different illumination conditions in Fig. 11. Specifically, normalized global illumination intensities of $[0, 0.025]$, $[0.1, 0.15]$, and $[0.2, 0.4]$ are adopted for statistical analysis. It can be seen that different combinations of experts are specialized in handling scenarios with different illumination intensities, validating the capability of the proposed model to adaptively handle degradation.

5.3. Discussion on model complexity and efficiency

In Table 7, we report the trainable parameters (denoted as "Params.(M)"), running time (s), giga floating-point operations per second (GFLOPs) cost, and peak memory usage (peak allocated memory for tensors on GPU while inferencing) of some representative SOTA models for LLE (i.e., FourLLIE, LLFormer, and UHDFour), LLE-Deblur (i.e., NAFNet, PDHAT, and VQCNIR), and LLE-SR (i.e., HAT, Restormer, and MIRNet-V2) tasks, respectively. As can be seen, the model size (calculated from learnable parameters) and running time of our model remain at a moderate level. In the enhancement stage, multi-scale feature extraction and large-size feature computation facilitate detail preservation, and also lead to relatively higher GFLOPs. It is noteworthy that our method performs well in terms of peak memory usage, facilitating the deployment in resource-constrained environments.

Table 7
Comparison on the trainable parameters, running time, computational cost, and peak memory usage.

Methods	Params(M)	Time(s)	GFLOPs	Mem.(MB)
FourLLIE	0.18	0.0169	2.86	171.07
LLFormer	24.55	0.0802	39.61	345.05
UHDFour	15.90	0.0843	57.42	394.42
IPCMoE-L (Ours)	5.65	0.0201	282.40	272.41
NAFNet	67.89	0.0328	63.30	482.48
PDHAT	7.83	0.0456	208.19	216.10
VQCNIR	45.90	0.0579	165.01	807.02
IPCMoE-D (Ours)	11.18	0.0407	559.75	340.85
HAT	20.51	0.2366	1383.20	3595.21
MIRNet-V2	5.96	0.0289	147.36	283.10
Restormer	26.24	0.0371	148.75	677.74
IPCMoE-S (Ours)	11.29	0.0390	566.66	341.99

5.4. Discussion on extremely challenging cases

In this section, we further explore the performance of the proposed method in extreme degradation scenarios (i.e., complex illumination conditions, blurry, and noise). Fig. 12 shows the results for dealing with extreme degradations. In Scene-A, the input image suffers from extremely low light, blurry, and noise. Our method effectively enhances brightness and restores the letters in the image. Scene-B exhibits severe motion blur, low light, and noise. None of the compared methods accurately recovers texture information, making this a failed case. In Scene-C, in addition to low light and blur, the noise degradation impacts detail cues. Consequently, the enhanced images produced by VQCNIR and the proposed method appear relatively clean, but the texture is not fully restored. In Scene-D and Scene-E, we showcase the nighttime image with artificial light sources and blurry. It can be observed that our method handles the scenario in Scene-D effectively and outperforms other state-of-the-art approaches. However, when dealing with large overexposed regions and severe texture loss (Scene-E), all methods fail to produce high-quality results.

5.5. Discussion on practical challenges and future works

This paper focuses on the low-light image restoration task, which plays an essential role in real-world applications such as

Fig. 12. The results of extremely challenging scenarios.

surveillance cameras, nighttime driving assistance, and intelligent transportation systems. While the proposed IPCMoE framework demonstrates promising results in handling complex degradations, several challenges remain for practical deployment. These include ensuring real-time inference on resource-limited hardware, achieving robust generalization under diverse illumination and degradation conditions, and maintaining computational efficiency without compromising restoration quality.

Looking forward, future work will explore light-weight expert architectures, model compression, and hardware-aware optimization to improve processing efficiency while maintaining high-quality restoration. For example, structured expert pruning, expert-level knowledge distillation, and low-bit quantization strategies will be investigated to further reduce computational and memory costs. Moreover, domain adaptation and self-supervised learning strategies will be investigated to enhance model robustness across diverse lighting conditions and imaging devices.

Beyond low-light restoration, the proposed Mixture-of-Experts design provides a flexible and general framework that can inspire future studies in multi-degraded image restoration, cross-modal enhancement, and unified visual perception tasks. Future research will focus on developing more efficient and interpretable MoE structures, and on integrating MoE with other advanced image restoration paradigms.

6. Conclusion

In this paper, we propose to integrate perceptual cues with mixture-of-experts for low-light image restoration. To this end, we design perceptual-integrated MoEs by developing customized routers and task-depended experts. Specifically, the T-MoE is developed to preserve valuable features to restore high-fidelity details, and the E-MoE that integrates perceptual cues is designed to formulate the relationship between feature enlightening and texture restoration. Extensive experiments demonstrate the superior performance of our method on SID, SDSD, LOL-Blur, Real-LOL-Blur, and RELLISUR datasets.

In addition, we consider two main challenges that remain for practical deployment: 1) real-time inference on resource-limited devices; 2) achieving robust generalization under diverse degradation conditions. Correspondingly, our future work will focus on light-weight model architectures, domain adaptation, and self-supervised learning strategies to further reduce computational costs while improving performance.

CRediT authorship contribution statement

Yuezhou Li: Writing – original draft, Methodology, Conceptualization; **Yuzhen Niu:** Funding acquisition, Conceptualization; **Huangbiao Xu:** Writing – review & editing, Investigation; **Rui Xu:** Writing – review & editing, Methodology; **Yuqi Hu:** Writing – review & editing, Validation; **Hui Da:** Methodology, Conceptualization; **Wenxi Liu:** Supervision, Conceptualization; **Lifang Wei:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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